Masoud Vahabzadeh

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Google Scholar
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Summary

- **Spatial Statistics & Remote Sensing Specialist** with 4+ years of experience in processing satellite-based datasets, spatiotemporal data fusion, and hydro-climatic modeling using Python and Google Earth Engine (GEE).
- **Water Resources & Environmental Systems Modeler** with 5+ years of experience in coupled hydro-environmental systems, resulting in 8 peer-reviewed journal publications with 300+ citations (h-index: 7).
- **Data Integration Expert** with practical experience in addressing spatial and temporal misalignments in environmental monitoring, aligning with nature-inclusive solutions**.**
- **Research Coordinator & Educator** experienced in managing collaborative research activities, teaching GEE for agricultural applications, and delivering 21 peer reviews for premier journals.

Education

M.Sc. in Civil Engineering (Water Resource Management), Iran University of Science and Technology- Tehran

GPA: 15.5 (out of 20) 2019 - 2022

Thesis title: Proposing a Holistic Framework for Energy Subsystem Simulation in The Context of The Water, Energy, and Food nexus System.

B.Sc. in Civil Engineering, University of Mohaghegh Ardabili- Ardabil 2014 - 2018

GPA: 15.46 (out of 20)

Research Interests

- Spatial and Spatiotemporal Modelling
- AI & Machine Learning Applications in Hydro-Environmental Systems
- Satellite Remote Sensing & Big Geospatial Data Analytics (GEE & Python)
- Surface and groundwater hydrology
- Satellite Remote Sensing & Geospatial Big Data Analytics
- Climate Change Impact Assessment & Adaptation Pathways

- Complex Coupled Systems Simulation & Optimization (WEF-Nexus)
- Machine Learning in Digital Agriculture & Hydrology
- River Discharge Estimation by Remote Sensing
- Hydro-Climatic Modeling & Precipitation Evaluation in Data-Scarce Basins

Work Experience

Research Assistant 2018 - 2022

Iran University of Science and Technology, Iran

- **Multidisciplinary Team Collaboration**: Co-led a diverse research team comprising one Ph.D. and three M.Sc. researchers to develop and execute comprehensive simulation-optimization models for Water-Energy-Food (WEF) Nexus systems.
- **Graduate Student Mentoring**: Served as the lead technical consultant for multiple M.Sc. thesis projects, providing expert advising and hands-on guidance on remote sensing methodologies, spatial data analytics, and Google Earth Engine (GEE) implementations.
- **Project Coordination & Team Science**: Coordinated complex research workflows and facilitated regular knowledge-sharing sessions, fostering a highly collaborative "Team Science" environment to meet critical research milestones.
- **Scientific Dissemination**: Co-authored and contributed to the development of high-impact research papers, resulting in publications in premier peer-reviewed outlets such as the Journal of Cleaner Production (2023).

Teaching Assistant 2022 - Present

Iran University of Science and Technology, Iran

- Passionate about knowledge dissemination and developing educational materials for students in geospatial analysis and remote sensing.
- Teaching ENVI software for agricultural project s.
- Taught Google Earth Engine for agricultural and water resource applications, guiding students in implementing advanced projects based on research published in high-impact journals (e.g., Remote Sensing of Environment, Science of The Total Environment, and Journal of Hydrology).

Technical and Research Experience

Reviewer for the following journals: 2022 - Present

- Journal of Cleaner Production (Reviewer of 3 papers)
- Energy (Reviewer of 2 paper)

- Applied Energy (Reviewer of 1 paper)
- Journal of Hydrology (Reviewer of 1 paper)
- Science of the Total Environment (Reviewer of 1 paper)
- Frontiers in Water (Reviewer of 1 paper)
- Frontiers in Environmental Science (Reviewer of 1 paper)
- Frontiers in Ecology and Evolution | Urban Ecology (Reviewer of 2 papers)
- Stochastic Environmental Research and Risk Assessment (Reviewer of 1 paper)
- Advances in Civil Engineering (Reviewer of 3 papers)
- Scientific Reports (Reviewer of 3 papers)
- Communications Earth and Environment (Reviewer of 1 paper)
- Discover Sustainability (Reviewer of 1 paper)

Selected Publications

Journal Papers:

1. **M. Vahabzadeh**, A. Afshar, A. Molajou, K Parnoon, SM Ashrafi, "A Comprehensive Energy Simulation Model for Energy-Water-Food Nexus System Analysis: A case study of the Great Karun Water Resources System ", *Journal of Cleaner Production*, (2023). 10.1016/j.jclepro.2023.137977. (Citation:31)

2. H, Akbari Variani, A. Afshar, **M. Vahabzadeh**, A. Molajou, MM, Akbari Varyani, "Development of a Novel Framework for Agriculture Simulation Model for Food-Energy-Water Nexus Analysis in Watershed-scale ", *Journal of Cleaner Production*, (2023). 10.1016/j.jclepro.2023.139492. (Citation:45)

3. **M. Vahabzadeh**, A. Afshar, A. Molajou, "Energy simulation modeling for water-energy-food nexus system: a systematic review", *Environmental Science and Pollution Research* (2023). https://doi.org/10.1007/s11356-022-24300-1. (Citation:47)

4. **M. Vahabzadeh**, A. Afshar, A. Molajou, "Framing a novel holistic energy subsystem structure for water-energy-food nexus based on existing literature (basic concepts) ", *Scientific Reports* (2023). https://doi.org/10.1038/s41598-023-33385-8. (Citation:40)

5. A. Afshar, E. Soleimanian, H A. Variani, **M. Vahabzadeh**, A. Molajou, "The conceptual framework to determine interrelations and interactions for holistic Water, Energy, and Food Nexus ", *Environment, Development and Sustainability* (2022). https://doi.org/10.1007/s10668-021-01858-3. (Citation: 113)

6. Akbari Variani, H., Afshar, A., **Vahabzadeh, M**. *et al.* A review on food subsystem simulation models for the water-food-energy nexus: development perspective. *Environ Sci Pollut Res* (2023). https://doi.org/10.1007/s11356-023-29149-6. (Citation: 23)

7. **M. Vahabzadeh**, A. Molajou, H, Akbari Variani, A. Afshar, "Assessment of water-energy nexus for the supply side of the energy subsystem for the water-stressed nation: Iran country, "Journal *of Cleaner Production*, (2025). https://doi.org/10.1038/s41598-025-21380-0. (Citation: 5)

8. Mohammad Reza Nikoo, Moein Tosan, Rouzbeh Nazari, **Masoud Vahabzadeh**, Maryam Karimi, Ehsan Zarrinabadi, "Assessing the fidelity of multi-satellite precipitation estimates for drought monitoring in a mountain water tower to arid basin system", "Journal of Arid Environments, (2026). https://doi.org/10.1016/j.jaridenv.2025.105519. (Citation: 9)

Conference Papers:

F. Rezaei Mogaddam, **M. Vahabzadeh**, *"Assessment and Determination of Vehicle Maneuvering Speeds on Curved Routes Using Trucksim Software," The First National Conference on Civil Engineering and Transportation (2017).* (During B.Sc)

Projects

- Comprehensive WEF-Nexus Simulation-Optimization (Teamwork)
  - **Challenge**: The lack of a holistic framework to model the complex interdependencies between Water, Energy, and Food (WEF) sectors.
  - **Solution**: Collaborated in a multi-disciplinary team to develop an integrated simulation model, leading to multiple peer-reviewed publications.
  - **Impact**: Established a dynamic framework for evaluating sustainable development policies and resource allocation under various scenarios.
- Evapotranspiration (ET) Mapping for Water Resource Management
  - **Challenge**: Lack of continuous spatial data for actual evapotranspiration, which is vital for the WEF-Nexus system.
  - **Solution**: Utilized GEE and ENVI to implement the SEBAL and OPTRAM-ET algorithms, integrating remote sensing data with meteorological parameters.
  - **Impact**: Facilitated a more precise estimation of water consumption in agricultural zones, supporting sustainable water management scenarios.
- Climate-Phenology-Hydrology Interaction Modeling
  - **Challenge**: Understanding the complex interactions between climatic variables and vegetation dynamics (phenology) in data-scarce regions.
  - **Solution**: Developed customized scripts utilizing the Google Earth Engine (GEE) API via Python to process and analyze massive multi-temporal geospatial datasets.
  - **Impact**: Provided actionable insights into how environmental stressors impact crop patterns and agricultural yield estimation.
- High-Resolution Soil Moisture Estimation under Uncertainty
  - **Challenge**: Mapping soil moisture dynamics across varied topographies while managing data uncertainties and outliers.

    - **Solution**: Implemented and calibrated the OPTRAM and TOTRAM algorithms using Google Earth Engine to estimate Surface Soil Moisture (SSM).
    - **Impact**: Delivered robust geostatistical outputs crucial for evaluating agricultural water demand and drought monitoring.
- River Discharge Estimation in Data-Scarce Basins
    - **Challenge**: Estimating hydrological flows in ungauged or poorly gauged river basins.
    - **Solution**: Applied the CM/CMW Method (analyzing dry and wet pixels) through automated Google Earth Engine workflows to monitor discharge variations.
    - **Impact**: Created a reliable alternative for continuous river monitoring using purely remote sensing data.
- Crop Pattern Classification and Rice Mapping
    - **Challenge**: Accurately identifying and differentiating crop types over large, heterogeneous agricultural areas is critical for food security and resource allocation.
    - **Solution**: Leveraged Google Earth Engine to process multi-temporal satellite imagery, developing classification models to map diverse crop patterns with a specific focus on rice cultivation.
    - **Impact**: Delivered high-resolution crop distribution maps that support sustainable agricultural planning and precision farming strategies.
- Bathymetry and Water Quality (Chlorophyll) Assessment
    - **Challenge**: Traditional monitoring of shallow water depths (Case 1 water) is cost-prohibitive and spatially limited.
    - **Solution**: Employed multispectral satellite data within Google Earth Engine to simultaneously model bathymetry (for 0-15m depths) and monitor chlorophyll concentrations.
    - **Impact**: Enabled cost-effective, large-scale monitoring of aquatic ecosystem health and physical conditions.
- Agricultural Yield Estimation
    - **Challenge**: Predicting regional crop yields requires processing massive volumes of geospatial, climatic, and phenological data efficiently.
    - **Solution**: Developed scalable, automated workflows using the Google Earth Engine API via Python to integrate remote sensing data for yield prediction.
    - **Impact**: Improved the accuracy and computational efficiency of crop production forecasting, providing vital data for agricultural management.
- Reservoir Water Quality Estimation via Machine Learning and Remote Sensing
    - **Challenge**: Overcoming the spatial and temporal limitations of traditional, point-based water quality sampling in large reservoirs to achieve continuous monitoring.
    - **Solution**: Implemented and evaluated Machine Learning algorithms applied to satellite imagery to estimate and predict key water quality variables.

- **Impact**: Established a scalable and cost-effective framework for the continuous spatiotemporal mapping of reservoir water quality, significantly enhancing water resource management and monitoring capabilities.

Computer Skills

- Programming & Scripting: Python (Data Analysis, Geostatistics, Spatiotemporal Modeling).
- Geospatial & Earth Observation: Google Earth Engine (GEE), ENVI.
- Data Handling: Handling multi-temporal and multi-spatial datasets, managing data sparsity and outliers.
- Acquainted with Machine Leaning
- Expert in Microsoft Office Word, Excel and PowerPoint

Certifications

- Multi-Objective Optimization with Python Bootcamp A-Z- Udemy Academy (2023)
- Optimization with Genetic Algorithms: Hands-on Python (2023)
- Optimization with GAMS: Operations Research Bootcamp A-Z (2023)
- Complete PYTHON Programming for Beginners - Udemy Academy (2021)

Hobbies

Sport (Soccer), Hiking, Cycling, Studying self-help, Novels, and psychological books

Language

English (Advanced)

References

References available upon request